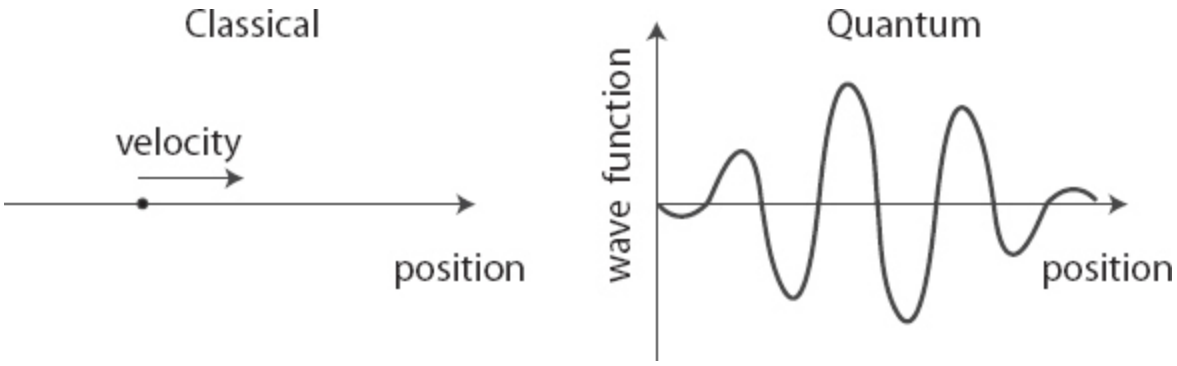


The probability of getting that outcome when we perform a measurement is given by the amplitude squared.

$$\text{Probability of a particular outcome} = |\text{Amplitude for that outcome}|^2$$

This simple relation is called the *Born rule*, after physicist Max Born.* Part of our task will be to figure out where in the world such a rule came from.

We're most definitely *not* saying that there is an electron with some position and velocity, and we just don't know what those are, so the wave function encapsulates our ignorance about those quantities. In this chapter we're not saying anything at all about what "is," only what we observe. In chapters to come, I will pound the table and insist that the wave function is the sum total of reality, and ideas such as the position or the velocity of the electron are merely things we can measure. But not everyone sees things that way, and for the moment we are choosing to don a mask of impartiality.

Let's place the rules of classical and quantum mechanics side by side to compare them. The state of a classical system is given by the position and velocity of each of its moving parts. To follow its evolution, we imagine something like the following procedure:

Rules of Classical Mechanics